

PEAK ACHIEVERS
ONLINE ACADEMY



PHYSICS

SENIOR SCHOOL

GRADE 10

BOOK 2
REVISION & EXAMINATION PRACTICE

- ✓ Comprehensive Revision
- ✓ Practical Activities
- ✓ Application Questions
- ✓ Mixed Revision
- ✓ 5 Examination Practice Tests (400 Marks)
- ✓ Complete Answers & Marking Guidance
- ✓ Rapid Revision & Formula Reference



**Competency-
Based Education
(CBE)**

Hezekiah Omweri Nyatuka
B.Ed. (Science), Kenyatta University
Chemistry & Physics Educator

*"You think Physics is very easy?
No; it is easier than that!"*

DEDICATION

This book is lovingly dedicated to the memory of my late father, Simeon Nyatuka, whose life, guidance and influence remain part of my journey.

To my mother, Hyrine Kerubo, for her love, strength, sacrifice and continued support.

To my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe, for their guidance and contribution to my growth.

To my friends - Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and many others.

To her, my special one; to my fellow teachers; and to every learner who continues working towards greater achievement.

- Hezekiah Omweri Nyatuka

PREFACE

Physics is not simply a collection of equations to memorise. It is a way of understanding motion, forces, energy, waves, electricity, materials, climate, technology and the universe around us.

Book 2 strengthens the same Grade 10 competencies through fresh examples, different applications, new practical situations and different revision and examination questions.

- Conceptual understanding
- Practical investigation
- Real-life application
- Numerical problem solving
- Scientific reasoning
- Environmental awareness
- Technology and examination preparation

PEAK ACHIEVERS PHYSICS SLOGAN: You think Physics is very easy? No; it is easier than that!

HOW TO USE THIS BOOK

- Read the brief notes before memorising equations.
- Study labelled diagrams and experimental arrangements.
- Follow worked examples: formula $\rightarrow$ substitution $\rightarrow$ calculation $\rightarrow$ unit.
- Attempt every Quick Check before reading its answer.
- Complete chapter revision questions independently.
- Attempt the Final Mixed Revision across all 13 chapters.
- Sit each 80-mark practice paper in 2 hours.
- Use the marking guidance to correct your reasoning, not merely copy answers.
- Finish with the Rapid Revision and Formula Reference section.

CONTENTS

1. Introduction to Physics
2. Pressure
3. Mechanical Properties of Materials
4. Temperature and Thermal Expansion
5. Moments and Equilibrium
6. Energy, Work, Power and Machines
7. Properties of Waves
8. Radioactivity and Stability of Isotopes
9. Electrostatics
10. Current Electricity
11. Introduction to Electronics
12. Greenhouse Effect and Climate Change
13. Introduction to Space Physics

Final Mixed Revision

National Examination Practice Tests 1-5

Answers & Marking Guidance

Rapid Revision & Formula Reference

About the Author

About Peak Achievers Online Academy

Final Learner Message

CHAPTER 1 - INTRODUCTION TO PHYSICS

Key Ideas

- Physics studies matter, energy, forces, motion and their interactions.
- Major areas include mechanics, thermal physics, waves, optics, electricity, electronics, nuclear physics and space physics.
- Scientific investigations commonly follow: problem -> question -> hypothesis -> experiment -> data -> analysis -> conclusion.
- Independent variable: deliberately changed. Dependent variable: measured. Controlled variables: kept constant.
- A measurement consists of a numerical value and a unit. Repeated readings and averaging improve reliability.

Formula Focus

$$\text{Mean} = \frac{\text{sum of readings}}{\text{number of readings}}$$

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

Worked Example

A pendulum gives periods 1.8 s, 1.9 s, 1.8 s and 1.7 s. Mean period = 1.8 s.

REMEMBER: Write the formula first, use SI units where required, show substitution clearly and include the final unit.

Quick Check

1. Why are repeated measurements useful?
2. A cyclist travels 360 m in 24 s. Calculate speed.

Quick Check Answers

1. They reduce the influence of random error and allow a mean to be found.
2. 15 m/s.

Application / Practical

Design a ruler reaction-time investigation. Identify the independent, dependent and controlled variables.

Chapter Revision Questions

1. Define Physics.
2. Distinguish independent and dependent variables.

3. State two reasons for repeating measurements.
4. A rod is measured as 52.4 cm, 52.6 cm and 52.5 cm. Find the mean.
5. A cyclist travels 360 m in 24 s. Find the speed.
6. Outline a fair investigation of how ramp height affects toy-car travel distance.

FREE SAMPLE - NOT FOR RESALE

CHAPTER 2 - PRESSURE

Key Ideas

- Pressure is normal force per unit area.
- For the same force, a smaller area produces greater pressure.
- Liquid pressure increases with depth and density and acts in all directions.
- Pascal's principle: pressure applied to an enclosed fluid is transmitted throughout the fluid.
- Atmospheric pressure explains the operation of straws, syringes and suction devices.

Formula Focus

$$P = F/A$$

$$P = \rho g h$$

$$F_1/A_1 = F_2/A_2$$

Worked Example

A 480 N crate rests on a 0.80 m x 0.30 m base. $P = 480/0.24 = 2000 \text{ Pa}$.

At 6 m in fresh water: $P = 1000 \times 10 \times 6 = 60,000 \text{ Pa}$.

REMEMBER: Write the formula first, use SI units where required, show substitution clearly and include the final unit.

Quick Check

1. Why do sharp knives cut more easily?
2. What happens to liquid pressure as depth increases?

Quick Check Answers

1. A sharp edge has smaller area, producing greater pressure for the same force.
2. It increases.

Application / Practical

Make holes at different heights in a water-filled bottle and compare the horizontal range of the jets.

Chapter Revision Questions

1. Calculate pressure when 450 N acts on 0.15 m^2 .
2. Explain why dams are thicker near the bottom.
3. State Pascal's principle.

4. A hydraulic system has $A_1=0.015 \text{ m}^2$, $A_2=0.45 \text{ m}^2$ and $F_1=180 \text{ N}$. Find F_2 .
5. Explain why a raised water tank improves pressure at taps.

FREE SAMPLE - NOT FOR RESALE

Application / Practical

Estimate human power by timing a learner climbing a measured vertical height.

Chapter Revision Questions

1. Calculate work done by 75 N over 4 m.
2. Find the power when 18,000 J is transferred in 30 s.
3. A machine lifts 640 N using 160 N. Find MA.
4. If $MA=3.6$ and $VR=4.5$, find efficiency.
5. Explain energy changes in a solar-powered water pump.

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CHAPTER 7 - PROPERTIES OF WAVES

Key Ideas

- Waves transfer energy without overall transfer of matter.
- Transverse vibrations are perpendicular to travel; longitudinal vibrations are parallel.
- Frequency is oscillations per second; period is time for one oscillation.
- Reflection obeys angle of incidence = angle of reflection.
- During refraction, speed and wavelength may change while frequency remains constant.
- Diffraction is spreading around obstacles or through openings.
- Sound needs a medium; electromagnetic waves can travel through vacuum.

Formula Focus

$$f = N/t$$

$$T = 1/f$$

$$v = f \lambda$$

$$\text{echo distance} = vt/2$$

Worked Example

150 oscillations in 6 s gives 25 Hz.

An echo after 0.60 s at 340 m/s gives distance 102 m.

REMEMBER: Write the formula first, use SI units where required, show substitution clearly and include the final unit.

Quick Check

1. A wave has $f=5$ Hz and $\lambda=0.80$ m. Find speed.
2. What happens to frequency when water waves enter shallow water?

Quick Check Answers

1. 4.0 m/s.
2. It remains constant.

Application / Practical

Use a ripple tank or tray to observe reflection, refraction or diffraction.

Chapter Revision Questions

1. Distinguish transverse and longitudinal waves.

NATIONAL EXAMINATION PRACTICE TEST 1

Time: 2 Hours Total: 80 Marks Take $g = 10 \text{ N/kg}$ where required.

SECTION A - 20 MARKS

1. Repeated measurements improve _____. [1]
2. A 600 N force acts on 0.25 m^2 . Find pressure. [2]
3. A spring length changes from 16.0 cm to 19.5 cm. Find extension in metres. [2]
4. Why are expansion gaps used in bridges? [1]
5. Find moment of 40 N at 0.25 m. [2]
6. A machine gives 2000 J useful output from 2500 J input. Find efficiency. [2]
7. A wave travels at 18 m/s and 6 Hz. Find wavelength. [2]
8. Find neutrons in sodium-23, $Z=11$. [1]
9. What happens between two positive charges? [1]
10. Find charge moved by 2 A in 15 s. [2]
11. Name a temperature-sensitive resistor. [1]
12. State one human activity that can increase atmospheric CO_2 . [1]
13. Name the galaxy containing the Solar System. [1]
14. State one use of an artificial satellite. [1]

SECTION B - 30 MARKS

15. A diver is 8 m below fresh water ($\rho=1000 \text{ kg/m}^3$). Calculate liquid pressure and explain why pressure is greater at greater depth. [6]
16. A 180 N force acts 0.80 m from a pivot. Find the balancing force at 0.60 m and state equilibrium conditions. [6]
17. A source has 3200 Bq and half-life 4 h. Find activity at 4 h, 8 h and 12 h; define half-life and state one safety measure. [6]
18. 4 ohm and 8 ohm resistors are in series across 12 V. Find total resistance, current and voltage across 8 ohm; state ammeter connection. [6]
19. Water waves have $f=5 \text{ Hz}$ and $\lambda=0.60 \text{ m}$. Find speed and describe changes on entering shallow water. [6]

SECTION C - 30 MARKS

20. Spring investigation: loads 0,2,4,6,8,10 N produce extensions 0,1,2,3,4,5 cm. Describe relationship, find k , state graph form, a control variable, elastic-limit meaning, one application and one reliability improvement. [10]

21. A greenhouse uses a thermistor, transistor and fan. Explain sensing/control; find fan power at 24 V and 2 A; state one renewable-energy advantage. [10]

22. A satellite monitors climate. Explain two observations, why EM waves work in space, find signal time for 6×10^7 m, state one hazard-warning use, explain enhanced greenhouse effect and give one renewable response. [10]

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GRADE 10 PHYSICS - BOOK 2

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